

A1 production kernel manifest – parameter-consistency gates (current config is a mock)

TECT verification-first repository · claim A1-PRODUCTION-KERNEL-MANIFEST

first issued 2026-06-23 · this version issued 2026-06-23 · v1.0

1. Purpose and scope

A production kernel config is admissible (and may be cited as the numerical implementation of A2/A3) only if it lies in the analytic branch (A1-SCALAR-ANALYTIC-BRANCH) AND its stored parameters are mutually consistent. This card states those gates and records that the CURRENT solver-template config fails them. OPEN until a certified config passes. Verification: [codes/foundations/a1_kernel_checks.py](#) (4/4 PASS, including the failure demonstration).

2. The manifest gates

For a config $(\mathbf{r}, Z, Y, q_0, \mu_2)$ with tolerance ε_{par} :

$$|Z + 2Y q_\star^2| \leq \varepsilon_{\text{par}}, \quad |r - (m_{\text{sh}}^2 + Y q_\star^4)| \leq \varepsilon_{\text{par}}, \quad |\mu_2 - (r - \frac{Z^2}{4Y})| \leq \varepsilon_{\text{par}}, \quad (1)$$

together with the branch condition $m_{\text{sh}}^2 = r - \frac{Z^2}{4Y} > 0$. The first gate enforces the shell $q_\star^2 = -Z/(2Y) = q_0^2$; the third enforces that the config's μ_2 names the SHELL mass m_{sh}^2 , not r .

3. The current config is a mock (FAILS)

For the solver template $(r, Z, Y, q_0) = (0.35, -1, 0.50, 0.6801747616)$ with $\mu_2 = 0.35$:

$$q_\star^2 = -\frac{Z}{2Y} = 1 \neq q_0^2 \simeq 0.4626 \quad (|Z + 2Y q_0^2| = 0.537), \quad m_{\text{sh}}^2 = r - \frac{Z^2}{4Y} = -0.15 < 0, \quad (2)$$

and $\mu_2 = 0.35 = r$ is the ZERO-momentum alias ($|\mu_2 - m_{\text{sh}}^2| = 0.50$). The manifest FAILS on three gates (assert `production_manifest_mock_config_FAILS`). Therefore this file is a **working/mock** config; it is NOT an N-001 production manifest, and A2/A3 must NOT cite it as their numerical implementation.

4. Devil's-advocate

(α) “*The solver still runs.*” VALID — it runs, but on a kernel outside the analytic branch ($m_{\text{sh}}^2 < 0$); its output cannot be the numerical realisation of theorems that assume $m_{\text{sh}}^2 > 0$. (β) “*This weakens A2/A3.*” DISMISSED — A2/A3 are unchanged; this card only forbids citing an inconsistent config as their implementation, which is an honesty gain, not a weakening. (γ) “ ε_{par} *unspecified.*” VALID — ε_{par} is a per-publication tolerance; the current config fails by $O(0.5)$, far above any reasonable ε . Quantitative sanity check: $m_{\text{sh}}^2 = -0.15$; shell residual 0.537; μ_2 -shell gap 0.50.

5. Result footer

Result ID: A1-PRODUCTION-KERNEL-MANIFEST (OPEN)
Precise statement: A production config is admissible iff $|Z+2Y q^2|$, $|r-(m_{sh2}+Y q^4)|$, $|\mu_2-(r-Z^2/4Y)| \leq \epsilon_{par}$ and $m_{sh2} > 0$. The current solver template FAILS ($m_{sh2} = -0.15 < 0$; $q^2 = 1 \neq q_0^2 = 0.463$; $\mu_2 = r$ alias).
Scope: parameter-consistency gate for production kernels.
Dependencies: A1-KERNEL-IDENTITY; A1-SCALAR-ANALYTIC-BRANCH.
Evidence grade: ANALYTIC (gates) + EXECUTED (mock-failure demo).
Reproduction command: python codes/foundations/a1_kernel_checks.py
Expected output: 4/4 PASS (incl. the mock-config FAILS assertion).
Tier: T1 OPEN (no certified production config yet).
Falsifier: a config passing all gates with $m_{sh2} > 0$ closes the OPEN status for that config.
No-overclaim statement: The current mock config is NOT certified; A2/A3 are NOT numerically implemented by it.
Next required action: Produce + certify an N-001 production config ($m_{sh2} > 0$, all gates) before any Class-II / condensate solver run is cited as the A2/A3 implementation.